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BME 364
MECHANISMS OF ALLERGIC DISEASE
Lab Manual

Group I

Paracellular flux of FITC-dextran 4 kDa

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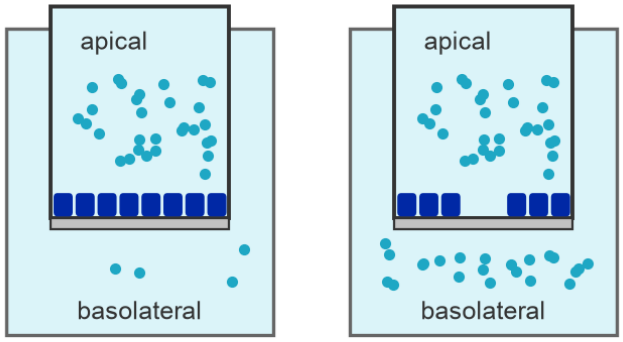
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Paracellular flux of FITC-dextran 4 kDa (FD4)

Objective: To measure the paracellular permeability of Caco-2 monolayers exposed to polysorbate 80 (P80; 0.02, 0.1 and 0.5 % v/v) from the apical-to-basolateral passage of FITC-dextran 4 kDa (FD4), expressed as the apparent permeability coefficient (P_{app}).



Control (little FD4 crosses) **P80 (more FD4 crosses)**

Donor (apical): 300 µL FD4, 0.5 mg/mL
Receiver (basolateral): 600 µL medium
Sample both sides after 24 h

$$P_{app} = \frac{C_{baso} \times V_{baso}}{t \times A \times C_0}$$

Materials

- The **Caco-2 Transwell inserts** of the TEER experiment (24-well plate: 4 inserts each of control, 0.02, 0.1 and 0.5 % P80, 1 blank), directly after the 24 h TEER reading.
- **FITC-dextran, average MW 4,000 (FD4):** light-sensitive, keep dark.
- Complete culture medium (DMEM + 10 % FBS + 1 % Pen/Strep) and the 0.02, 0.1 and 0.5 % P80 solutions (see the TEER manual).
- **Black 96-well plate** (flat bottom) for fluorescence.
- **Fluorescence microplate reader:** excitation ~490 nm, emission ~520 nm (e.g. 498/517 nm).
- 1.5 mL tubes, aluminium foil, multichannel pipette.

Solutions

FD4 Dosing Solutions (0.5 mg/mL = 500 µg/mL)

1. **Control FD4:** dissolve 2.5 mg FD4 in 5 mL complete medium (inserts + standards).
2. **P80 FD4, one tube per concentration:** dissolve 1 mg FD4 in 2 mL of the 0.02, 0.1 or 0.5 % P80 solution, so the P80 exposure continues until 48 h.
3. Vortex, wrap the tubes in foil and keep them at 37°C until use.

Standard Curve (prepare on the sampling day, 48 h)

1. Label 7 tubes: 25, 12.5, 6.25, 3.13, 1.56, 0.78 and 0.39 µg/mL, and one "BLK" tube.
2. Top standard: 50 µL of Control FD4 (500 µg/mL) + 950 µL complete medium = 1 mL of 25 µg/mL (1:20).
3. Add 500 µL of complete medium to the other six tubes and to BLK.
4. Two-fold serial dilution: transfer 500 µL from 25 to 12.5 µg/mL, mix, and continue down to 0.39 µg/mL.
5. Make standards and samples in the **same medium**; BLK is medium only.

Plate maps

24-well Transwell plate (the inserts)

The same inserts as in the TEER assay, one condition per row. The insert is named by its position (A1, B3, …), and the samples on the black plate carry these names.

	1	2	3	4	5	6
A	CTRL	CTRL	CTRL	CTRL	BLK	
B	0.02 %	0.02 %	0.02 %	0.02 %		
C	0.1 %	0.1 %	0.1 %	0.1 %		
D	0.5 %	0.5 %	0.5 %	0.5 %		

CTRL (A1–A4): Caco-2 insert, control FD4 in the apical compartment (0 % P80).

0.02 % (B1–B4), 0.1 % (C1–C4), 0.5 % (D1–D4): Caco-2 insert, FD4 with P80 at that concentration in the apical compartment.

BLK (A5): Insert without cells. It gets medium without FD4 and stays the TEER blank.

Black 96-well plate (the reading)

Black 96-well plate, 100 µL per well, every sample in duplicate.

	1	2	3	4	5	6	7	8	9	10	11	12
A	25	25	A1	A1	C1	C1	A1	A1	C1	C1		
B	12.5	12.5	A2	A2	C2	C2	A2	A2	C2	C2		
C	6.25	6.25	A3	A3	C3	C3	A3	A3	C3	C3		
D	3.13	3.13	A4	A4	C4	C4	A4	A4	C4	C4		
E	1.56	1.56	B1	B1	D1	D1	B1	B1	D1	D1		
F	0.78	0.78	B2	B2	D2	D2	B2	B2	D2	D2		
G	0.39	0.39	B3	B3	D3	D3	B3	B3	D3	D3		
H	BLK	BLK	B4	B4	D4	D4	B4	B4	D4	D4		

Columns 1–2, STD: standards in µg/mL (rows A–G) and BLK (row H, medium only).

Columns 3–6: basolateral samples, undiluted, in duplicate. Labels are the insert positions: A = control, B = 0.02 %, C = 0.1 %, D = 0.5 % P80.

Columns 7–10: apical samples diluted 1:50, same order.

Steps

FD4 Addition (directly after the 24 h TEER reading)

□ 1. Aspirate the basolateral medium of all inserts and add 600 µL of fresh, warm complete medium.

□ 2. Aspirate the apical medium with the tip at the insert wall. Add 300 µL of Control FD4 to A1–A4 and 300 µL of the matching P80 FD4 to B1–B4 (0.02 %), C1–C4 (0.1 %) and D1–D4 (0.5 %). The blank insert A5 gets 300 µL of medium without FD4 (it stays the TEER blank).

□ 3. Note the time, wrap the plate in aluminium foil and incubate for 24 hours at 37°C, 5 % CO_2.

Sampling (48 h, after the 48 h TEER reading)

□ 4. Prepare the standard curve.

□ 5. Basolateral: mix gently by pipetting up and down three times, then transfer 2 × 100 µL to the black plate (columns 3–6).

□ 6. Apical: mix gently, transfer 20 µL into 980 µL of medium (1:50), vortex, then transfer 2 × 100 µL to the plate (columns 7–10). Undiluted apical samples (~500 µg/mL) are far above the top standard.

□ 7. Add 2 × 100 µL of each standard and of BLK (columns 1–2).

□ 8. Read the plate immediately: excitation ~490 nm, emission ~520 nm, top reading. Set the gain on the 25 µg/mL standard so that it is not saturated, and keep the settings for the whole plate.

□ 9. Export the values with their well positions and fill in the two tables below.

conc_ug_ml	rfu
0	52
0	47
0.39	409
…	…
25	23401

insert	condition	side	dilution	replicate	rfu
A1	ctrl	basolateral	1	1	1087
A1	ctrl	basolateral	1	2	1103
A1	ctrl	apical	50	1	9852
A1	ctrl	apical	50	2	9716

Data analysis

Standard Curve and Concentrations

□ 10. Fit a straight line through the standards (RFU = a + b × concentration, 0.39–25 µg/mL). R^2 should be above 0.99.

□ 11. Check that every sample lies inside the standard range; otherwise dilute and read again (at 0.5 % P80 the basolateral samples come close to the top standard).

□ 12. Average the duplicate wells and convert RFU to concentration; multiply apical values by the dilution (50):
C (µg/mL) = (RFU − a) / b × dilution

Amount Crossed and Recovery

□ 13. Amount that crossed into the basolateral compartment, and the fraction of the dose (dose = 500 µg/mL × 0.3 mL = 150 µg):

$Q\ (\mu g) = C_{baso} \times V_{baso} = C_{baso} \times 0.6\ mL$
$\%\ of\ dose = Q / 150\ \mu g \times 100$

□ 14. Recovery = $(Q + C_{api} \times 0.3\ mL) / 150\ \mu g \times 100$. It should be close to 100 %; much less means FD4 was lost or sampled wrongly.

□ 15. Sink conditions: the transported fraction should stay below ~10 % of the dose so the gradient stays constant.

Apparent Permeability

□ 16. Calculate P_{app} (cm/s) for every insert:
$P_{app} = Q / (t \times A \times C_0)$
$t = 86{,}400$ s, $A = 0.33\ cm^2$, $C_0 = 500\ \mu g/mL\ (= \mu g/cm^3)$

Example: $C_{baso} = 1.12\ \mu g/mL \rightarrow Q = 1.12 \times 0.6 = 0.672\ \mu g \rightarrow P_{app} = 0.672 / (86{,}400 \times 0.33 \times 500) =$ **4.7×10^{-8} cm/s**

Statistical Analysis & Visualization

□ 17. P_{app} values spread over orders of magnitude: analyse $\log_{10} P_{app}$. Compare every P80 concentration with the control by one-way ANOVA followed by Dunnett's test (n = 4 inserts per condition); $10^{difference}$ is the fold change. Games–Howell when the SDs differ, Kruskal–Wallis + Dunn (Holm) as the rank-based check.

□ 18. Relate the two barrier read-outs: Spearman correlation of P_{app} with the 48 h TEER of the same inserts. Knitting **04_flux.Rmd** (folder R/mac_linux or R/windows) runs all steps (knit 03_teer.Rmd first).

□ 19. Plot P_{app} (log scale) per condition, or against TEER, one point per insert.

□ 20. Report the tracer (FD4), incubation time, area and C_0 with P_{app}, and read it together with TEER: TEER down with unchanged FD4 flux points to the pore (ion) pathway; more FD4 flux points to the leak pathway or lesions.

References

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